

Assignment-1

1. Consider an $M/M/1/1$ queue with two classes of jobs. Class $i \in \{1, 2\}$ jobs arrive according to independent Poisson processes with rates λ_i , and their service times are exponentially distributed with rate μ_i .
 - (a) Derive the long-run proportion of time the server is occupied by class i jobs, for $i = 1, 2$.
 - (b) Compute both the time-average and ensemble-average occupation probabilities for each class and verify whether they are equal or not
2. Consider a single-server queue with Poisson arrivals of rate λ . Service times are i.i.d. with general distribution G and mean $E[S]$. Whenever the system becomes empty, the server turns off immediately. Upon the next arrival to an empty system, the server requires a random setup time U (independent of arrivals and service times) before beginning service. Assume $E[U] < \infty$ and $\lambda E[S] < 1$.

Determine the long-run fraction of time the server is:

- (a) busy serving jobs,
- (b) in setup,
- (c) idle (powered off).

Refer to Isi mitrani section 3.2.1 for hint.

3. Consider a network of k servers, each with its own unbounded queue. Jobs at a server are processed in First-Come, First-Served (FCFS) order. The i -th server has an exponentially distributed service time with rate μ_i .

Each server may receive arrivals from both inside and outside the network. The arrivals into the i -th server from outside the network constitute a Poisson process with rate r_i .

The routing of jobs is probabilistic. Specifically, every job that completes service at server i will be transferred to server j with probability P_{ij} , or will exit the system with probability

$$P_{i,out} = 1 - \sum_{j=1}^k P_{ij}.$$

Find the stationary distribution for the above network.

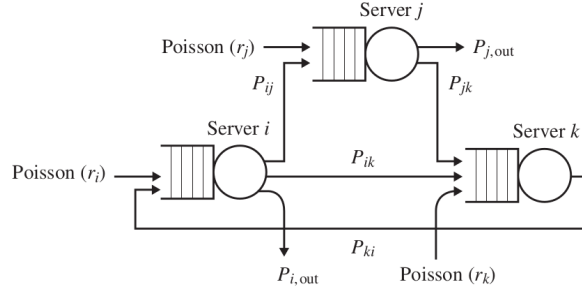


Figure 1: An example network with only some selected P_{ij} shown for cleanliness

4. Consider an $M/M/1$ queue with two classes of jobs. Class $i \in \{1, 2\}$ jobs arrive according to independent Poisson processes with rates λ_i , and service times are exponentially distributed (with common service rate μ). Define the state of the system at time t as the ordered sequence of job classes currently present in the system, including the job in service. For example, if there are three jobs in the system and the job in service is of type 1 while the two waiting jobs are of type 2, then the state is represented by $(1, 2, 2)$. Derive the stationary distribution of this Markov process.